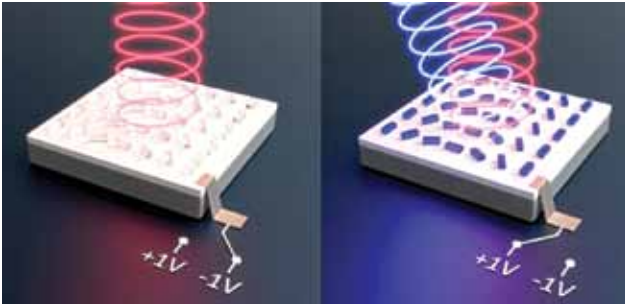


Nanoantennas enable holograms for virtual conferences

Researchers at the University of Stuttgart have succeeded in breaking the constraints related to liquid-crystal technology by developing electrically switchable plasmonic nanoantennas with dimensions of only a few hundred nanometers and made from conductive polymers. This will help in

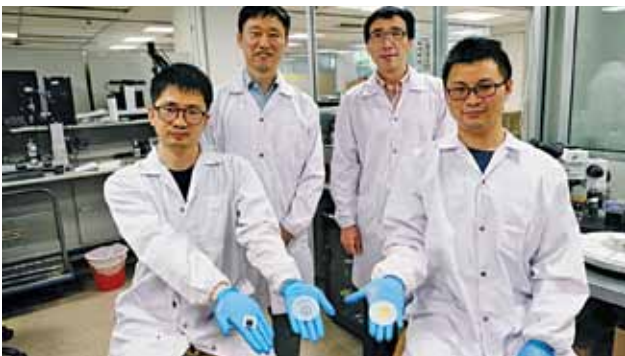


Left: Image displaying the plasmonic polymer nanoantenna, switched into the dielectric (glassy) state. The light beam from the bottom just passes through to the top without being deflected. Right: Image displaying the plasmonic polymer nanoantenna, switched into the metallic state. The light beam from the bottom is deflected to the side when passing through the sample. The light also changes its handedness when travelling to the top (Credit: www.uni-stuttgart.de/en)

realising dynamic holographic displays, which can enable virtual conferences with a 'real-life feeling. It was demonstrated that the optical appearance of the nanoantennas could be switched between that of a shiny metal and a transparent material by applying a voltage between minus and plus 1V at video rates of 30Hz. Despite being only a few tens of nanometers thick and less than 400 nanometers in size, the nanoantennas do the same job as the much larger and thicker liquid crystals used in current state-of-the-art technology.

Flexible, paper-thin biodegradable printed batteries

Nanyang Technological University, Singapore scientists have developed paper-thin biodegradable zinc batteries



Batteries of the future could be paper-thin and biodegradable (Credit: www.ntu.edu.sg)

that could one day become an environmentally sustainable option for powering flexible and wearable electronic systems. These zinc batteries are made up of electrodes screen-printed on to both sides of a piece of cellulose paper that has been reinforced with hydrogel. Once the battery has been expended, it can be buried in soil, where it breaks down completely within a month. The scientists think their printed battery could be integrated into flexible electronics, such as foldable smartphones or biomedical sensors, for health monitoring.

Non-invasive device can treat stroke and diabetes patients

A silicon device that can change skin tissue into blood vessels and nerve cells has been developed by researchers at the Indiana University School of Medicine, bringing us closer to potential use as a treatment for people with a variety of health concerns. The technology, called tissue



Silicon nanochip can reprogram biological tissue in living body (Credit: <https://news.iu.edu>)

nanotransfection, uses a non-invasive nanochip device that can reprogram tissue function by applying a harmless electric spark to deliver specific genes in a fraction of a second. It is suitable for reprogramming tissue for different kinds of therapies, such as repairing brain damage caused by stroke or preventing and reversing nerve damage caused by diabetes.

Smart bandage for onsite chronic wound monitoring

The research team from the National University of Singapore's (NUS) Department of Biomedical Engineering, in collaboration with Singapore General Hospital, has developed a smart wearable sensor that can conduct real-time, point-of-care assessment of chronic wounds wirelessly via an app. A world's first, the novel sensor technology can detect temperature, pH, bacteria type, and inflammatory factors specific to chronic wounds within fifteen minutes,